



SLIDE 1

(Q3 – Principles, Standards, Guidelines, Heuristics + Evaluation Methods)

♦ EXACT QUESTION (VERBATIM)

Q3:

(a) How would you differentiate between principles, standards, guidelines, and heuristics, using at least one example of each to illustrate? (2 marks)

(b) You have studied the following different evaluation methods:

- iv. Heuristic Evaluation
- v. Cognitive Walkthrough
- vi. Think Aloud
- vii. Usability Testing

In your opinion, which type of method is best used to evaluate different user groups and in what situations? Please add your justifications for selecting a method for a particular user group. The User Group is divided into elders, teenagers, disabled people, and illiterates. (3 marks)

♦ WHAT THIS QUESTION IS ASKING

This question tests **conceptual clarity + application**.

- **Part (a):** Can you clearly distinguish *similar-looking concepts*
- **Part (b):** Can you **match evaluation methods to user types** logically?

Your teacher wants:

- **Clear differentiation**
 - **Concrete examples**
 - **Justified choices**, not generic definitions
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◆ HOW TO THINK ABOUT THE ANSWER

Part (a)

You MUST:

- Define each term
- Say how it differs
- Give **ONE example EACH**

Part (b)

You MUST:

- Pick **best method per user group**
 - Explain **why** that method suits that group
 - Tie it to **cognitive ability, literacy, comfort level**
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✓ PERFECT EXAM ANSWER

(a) Differentiation

Principles are fundamental design beliefs that guide good design, such as *visibility* or *feedback*. For example, providing immediate feedback after a button click.

Standards are mandatory rules defined by organizations, such as ISO usability standards that must be followed.

Guidelines are recommended best practices that designers should follow when possible, such as maintaining consistency in layouts.

Heuristics are general usability rules used to evaluate interfaces, such as Nielsen's heuristic of *recognition rather than recall*.

(b) Evaluation methods for user groups

For **elderly users**, usability testing is most effective because it observes real interaction difficulties such as memory and motor issues.

For **teenagers**, think-aloud is suitable as they are comfortable verbalizing thoughts while interacting.

For **disabled users**, cognitive walkthrough helps evaluate accessibility barriers step by step.

For **illiterate users**, usability testing is best since it does not rely on verbal explanation or reading ability, but on observed behaviour.



SLIDE 2

(Heuristic Evaluation – Amazon Book Website)

◆ EXACT QUESTION (VERBATIM)

Q2:

The development manager of a website for online book-buying has asked you to carry out a heuristic evaluation of its usability. He has specifically proposed the three heuristics listed below:

1. “There should be between five and nine navigation options on each page.”
2. “There should be a good match between the navigation buttons and the users’ goals.”
3. “It should be easy for users to change their plans.”

(a) Has the manager misunderstood heuristic evaluation? Briefly justify your answer. *(2 marks)*

(b) Please comment on the above three heuristics suggested by the manager. For each of the proposed heuristics, your comments should include:

- i. Any theoretical justification for (or against) this heuristic;
 - ii. Any additional evaluation steps that might be required in applying it; and
 - iii. The likely impact of such evaluation on the system design. *(3 marks)*
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◆ WHAT THIS QUESTION IS ASKING

This is a **trap question**.

The manager:

- Treats heuristics as **rules**
- Not **evaluation principles**

Your job:

- Correct him
 - Show theory
 - Show evaluation impact
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◆ HOW TO THINK ABOUT THE ANSWER

- Part (a): One strong paragraph
 - Part (b): **Structure matters more than length**
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✓ PERFECT EXAM ANSWER

(a) Misunderstanding?

Yes, the manager has misunderstood heuristic evaluation. Heuristics are general usability principles meant to guide evaluation, not rigid rules such as fixed numerical limits.

(b) Comments

1. Five to nine navigation options

This is loosely based on cognitive load theory; however, enforcing a fixed number is inappropriate. Evaluation would require observing user navigation behaviour. This may result in restructuring menus based on user needs rather than arbitrary limits.

2. Match between navigation buttons and user goals

This aligns with Nielsen's heuristic of match between system and real world. Task-based testing would be required. The likely impact is improved task efficiency and reduced user confusion.

3. Easy to change plans

This reflects user control and freedom. Evaluation would require testing undo and flexibility features. The impact would be reduced frustration and increased user confidence.



SLIDE 3

(Supporting New Classes of Users – Banking Website)

♦ EXACT QUESTION (VERBATIM)

(c) After the project has begun, it is decided that the new design should also support two further classes of users: those who do not want to make a transaction online, but simply need to find out through their application, what services their branch can offer? And the second class is of the users who want to make telephonic calls from their application to the call-center to find out more about the services their physical branch can provide.

What design changes you will use to customize the user interface for these classes of users, and what improvements might they experience as a result? (2 marks)

♦ WHAT THIS QUESTION IS ASKING

They want:

- Customization
 - Non-transactional users
 - Reduced friction
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♦ HOW TO THINK ABOUT THE ANSWER

Mention:

- Service discovery
 - Call integration
 - Simplicity
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PERFECT EXAM ANSWER

For users who only want service information, I would add a dedicated “Branch Services” section with clear categories. For users who want to contact the call center, I would integrate a one-tap call button within the application. These changes would reduce user effort, improve accessibility, and increase satisfaction for non-transactional users.

SLIDE 4

(Dashboard Redesign – Coordination & Awareness)

◆ EXACT QUESTION (VERBATIM)

(a) You suspect that different users use the account detail page in several quite distinct ways. Instead of simply checking balance, they wish to use this page to do further transactions, such as payment of bills or online cash transfer, etc. Based on this reading, you feel that some design change of the dashboard page should be conducted to remove this problem. Secondly, you also want to incorporate the main aspects of social mechanisms of “coordination among pages” and “awareness of the users where they are and what they intend to do”.

What will be the basic design changes that you will incorporate on the dashboard page? State any two major changes. (2 marks)

◆ WHAT THIS QUESTION IS ASKING

- Dashboard ≠ static
- Users have **different intents**
- Apply **awareness + coordination**

PERFECT EXAM ANSWER

I would add clearly labeled shortcuts for common tasks such as bill payment and fund transfer directly on the dashboard. Additionally, I would include visual cues indicating the user's current location and available next actions. These changes improve coordination across pages and increase user awareness.



SLIDE 5

(User Behaviour Logs – Errors & Help)

◆ EXACT QUESTION (VERBATIM)

(b) A study of the system data logs shows two surprising patterns of behaviour. Some users repeatedly enter customer ID and Branch ID to use for transactions instead of their account number. The other pattern includes long pauses between entering each piece of data, with the same users accessing help and documentation pages in alternation with data entry pages. When they access help in the middle of the transaction process, the application takes them to the dashboard page after completion of the 'help' process.

- i. How will you handle these two patterns in your new design? (2 marks)
 - ii. What usability analysis procedures might you carry out to improve the usability of the system for these two classes of the user? (2 marks)
 - iii. What improvements might be experienced by each class as a result? (2 marks)
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◆ WHAT THIS QUESTION IS ASKING

They want:

- Error prevention
- Inline help
- Better flow continuity

PERFECT EXAM ANSWERS

i. Handling behaviour patterns

I would add input validation and contextual hints to guide users toward correct identifiers. I would also ensure that help pages return users to the exact step they were previously on.

ii. Usability analysis procedures

I would conduct usability testing to observe real interaction patterns and perform think-aloud sessions to understand user confusion during data entry.

iii. Improvements experienced

Users would experience fewer errors, reduced cognitive load, and smoother task completion without losing progress during help access.



SLIDE 6

(Memorability & Learning – Online Banking Interface)

◆ EXACT QUESTION (VERBATIM)

Assume you are preparing a new interface for an online banking application. The major problem the bank is facing is the memorability and learning of the software. Users have always complained that they forget how to use this application, despite being frequent users. Usability experts believe that the major problem is in the perception of the system to users. Many users have complained that they have to learn every new process and interaction, rather than performing the actual banking transaction, such as paying a utility bill or fund transfer.

As an expert, the bank has assigned you the task to improve the usability of their application. The following are the screenshots of the existing login page, dashboard, and account details page.

◆ WHAT THIS QUESTION IS ASKING

This is about **recognition vs recall**, **consistency**, and **mental models**.

◆ HOW TO THINK ABOUT THE ANSWER

Mention:

- Familiar patterns
 - Reduced learning
 - Task-first design
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✓ PERFECT EXAM ANSWER

To improve memorability, I would redesign the interface to prioritize common tasks such as bill payment and fund transfer using consistent icons and layouts. I would reduce unnecessary information on the account details page and ensure uniform navigation across screens. This would allow users to recognize actions instead of recalling steps, resulting in faster and more confident interaction.